Teo Gil Moreno de Mora Sardà

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Professional Experience

2025 – 2026 **Research and Teaching Assistant** (ATER), *Université Paris-Est Créteil*

2022 – 2025 **PhD student and Teaching Assistant**, *Université Paris-Est Créteil, Universitat Autònoma de Barcelona*

Education

2022 – 2025 **PhD in Mathematics**, *Université Paris-Est Créteil, Universitat Autònoma de Barcelona*, (cotutelle)
Thesis: A metric approach to the study of manifolds of positive scalar curvature.
Supervisors: Stéphane Sabourau, Florent Balacheff
Defense date: 29 September 2025

2021 – 2022 **Master in Fundamental Mathematics**, *Sorbonne Université, Université Paris Cité, Université Sorbonne Paris Nord, France*
Thesis: Filling volume minimality for metrics close to the Euclidean metric.
Supervisor: Stéphane Sabourau

2017 – 2021 **BSc in Mathematics**, *Universitat Autònoma de Barcelona, Spain*
Thesis: Generalizations of the first Minkowski's theorem from a systolic geometry perspective.
Supervisor: Florent Balacheff

2015 – 2021 **BSc in Physics**, *Universitat Autònoma de Barcelona, Spain*
Thesis: Dynamical Superfluid-Insulator Transition of Bose–Einstein condensates in optical lattices.
Supervisor: Verònica Ahufinger

Publications

[3] T. Gil Moreno de Mora Sardà, *Urysohn width of hypersurfaces and positive macroscopic scalar curvature*, preprint arXiv:2504.06737 (2025). Accepted for publication in J. Topol. Anal.

[2] F. Balacheff, T. Gil Moreno de Mora Sardà and S. Sabourau, *Complete 3-manifolds of positive scalar curvature with quadratic decay*. Math. Ann. 392, 4361–4389 (2025)

[1] F. Balacheff and T. Gil Moreno de Mora Sardà, *Isosystolic inequalities on two-dimensional Finsler tori*. EMS Surv. Math. Sci. 11, no. 2, 205–233 (2024)

Talks

Talks in conferences

Jan 2026 Congreso Bienal Real Sociedad Matemática Española, Sesión Análisis Geométrico, Alacant, Spain

Jul 2025 Convergence of Scalar Curvature 2025 Summer Workshop, *Online*

Jun 2025 Flows and Singular Spaces, *Centre Henri Lebesgue*, Nantes, France

Apr 2025 Rencontre de l'ANR OrbiScaR, *Université Paris-Est Créteil*, France

Mar 2025 Systolic and Diastolic Geometry, *Instituto de Matemática Pura e Aplicada*, Rio de Janeiro, Brazil

Nov 2024	Reunión de investigadores jóvenes de la REAG, *Universidade de Santiago de Compostela*, Spain
Oct 2024	Math SOMMa Junior Meeting, *Centre de Recerca Matemàtica*, Barcelona, Spain
Nov 2023	Forum des jeunes mathématicien·nes, *Université Libre de Bruxelles*, Belgium
May 2023	Journées des Doctorants du LAMA, *Université Gustave Eiffel and Université Paris-Est Créteil*, France
Mar 2020	Bernardo Cascales Functional Analysis School, *Universidad de La Laguna*, Spain
	Seminar talks
Jul 2025	Geometry Seminar, *University of Göttingen*, Germany
Jul 2025	Geometry Seminar, *Max Planck Institute*, Leipzig, Germany
Apr 2025	Séminaire de Théorie Spectrale et Géométrie, *Institut Fourier*, Grenoble, France
Feb 2025	Séminaire des Doctorants, *Institut Élie Cartan de Lorraine*, Nancy, France
Dec 2024	Séminaire Gaston Darboux, *Institut Montpelliérain Alexander Grothendieck*, Montpellier, France
Dec 2024	Geometry and Topology Seminar, *Université du Luxembourg*, Luxembourg
Dec 2024	Hodge Club, *University of Edinburgh*, United Kingdom
Oct 2024	Séminaire de Géométrie, *Institut de Mathématiques de Bordeaux*, France
Jun 2024	Seminari de Geometria, *Universitat Autònoma de Barcelona*, Spain
May 2024	Séminaire de Géométrie, *Institut de Mathématiques de Jussieu - Paris Rive Gauche*, France
Apr 2024	Colloque de Géométrie, *Université Paris-Est Créteil*, France
Apr 2024	Séminaire des doctorants du LAMA, *Université Paris-Est Créteil*, France
Jun 2023	Seminari Informal de Matemàtiques de Barcelona, *Universitat de Barcelona*, Spain
Mar 2023	Seminari de Geometria, *Universitat Autònoma de Barcelona*, Spain
	Poster presentations
Nov 2024	Curvature and Geometric Analysis in Rome, *Sapienza Università di Roma*, Italy
Jun 2024	European Doctorate School of Differential Geometry, *Instituto de Matemáticas de la Universidad de Granada*, Spain

Teaching

Jan 26-May 26	Fonctions de plusieurs variables (L1 Physique) *Université Paris-Est Créteil*, 18h Tutorials
Jan 26-May 26	Linear Algebra (L2 Économie et Gestion) *Université Paris-Est Créteil*, 72h Lectures and Tutorials
Sep 25–Dec 25	Analyse (L2 Économie et Gestion) *Université Paris-Est Créteil*, 36h Lectures and Tutorials
Sep 25–Dec 25	Calculus 1 (L1 Physique) *Université Paris-Est Créteil*, 30h Lectures and Tutorials
Jan 25–Apr 25	Analyse Réelle (L1 Mathématiques) *Université Paris-Est Créteil*, 36h Tutorials
Jan 24–Apr 24	Analyse Réelle (L1 Mathématiques) *Université Paris-Est Créteil*, 36h Tutorials
Sep 22–Dec 22	Calculus 1 (L1 Chimie) *Université Paris-Est Créteil*, 30h Lectures and Tutorials

Organization of Scientific Activities

May 2024	Journées des Jeunes Chercheurs·ses du LAMA, *Institut Henri Poincaré*, France (22-24 May 2024)
2022 – 2024	Séminaire des doctorants, *Laboratoire d'Analyse et Mathématiques Appliquées, Université Paris-Est Créteil*, France

Awards and Scholarships

2021 – 2022	**PGSM Master Scholarship**, granted by the *Fondation Sciences Mathématiques de Paris*.
2020 – 2021	**Beca de Colaboración grant BDNS 512590** by the *Spanish MEFP*
2015 – 2021	Academic Merit State Scholarship, funded 33% of my undergraduate degrees.
2015	Extraordinary University Entrance Examination Award, awarded by the *Generalitat de Catalunya*.
2015	University Entrance Scholarship, awarded by the *Fundació Catalunya La Pedrera*.

Languages

Catalan Native

Spanish Native

English Fluent
C2 Level - Cambridge Certificate

French Fluent
DELF B2

Japanese Intermediate
B1 Level - JLPT N3

Computer skills

Programming Languages Python, C, R

Other SageMath, LaTeX

Undergraduate Research Experience

Sep 20 –Jul 21 Research Assistant, Mathematics department UAB, *Barcelona, Spain*
Project: Systolic geometry of Finsler tori.
Supervisor: Florent Balacheff

Jul 20 –Sep 20 ICMAT Introduction to Research Program, *Madrid, Spain*
Project: Contact geometry and the space of light geodesics of a Lorentz manifold.
Supervisor: Francisco Presas

Jul 2019 IFISC Summer Undergraduate Research Internship, *Palma de Mallorca, Spain*
Project: Analysis of language variation in the United Kingdom using Twitter data and Big Data techniques.
Supervisor: David Sánchez

Jul 18 – Oct 18 ICFO Summer Fellows Program, *Barcelona, Spain*
Project: Development of an magnetic field active stabilisation system for the experimental set-up of the group.
Supervisor: Leticia Tarruell – Ultracold Quantum Gases Group

Jul 2016 ICFO Introductory internship, *Barcelona, Spain*
Project: Programming random walks on different geometrical settings.
Supervisor: Miguel Ángel García – Quantum Optics Theory Group

Attended conferences and schools

Oct 2025 Géométrie: échanges et perspectives, *Institut Henri Poincaré*, Paris, France

Jun 2025 Flows and Singular Spaces, *Centre Henri Lebesgue*, Nantes, France

Apr 2025 Rencontre de l'ANR OrbiScaR, *Université Paris-Est Créteil*, France

Mar 2025 Systolic and Diastolic Geometry, *Instituto de Matemática Pura e Aplicada*, Rio de Janeiro, Brazil

Jan 2025 Analysis and Geometry on Groups and Spaces Winter School, *Hausdorff Research Institute for Mathematics*, Bonn, Germany

Nov 2024 Reunión de investigadores jóvenes de la REAG, *Universidade de Santiago de Compostela*, Spain

Nov 2024 Curvature and Geometric Analysis in Rome, *Sapienza Università di Roma*, Italy

Oct 2024 Math SOMMa Junior Meeting, *Centre de Recerca Matemàtica*, Barcelona, Spain

Aug 2024 Metric Geometry and Geometric Measure Theory, *Université de Fribourg*, Switzerland

Jun 2024 European Doctorate School of Differential Geometry, *Instituto de Matemáticas de la Universidad de Granada*, Spain

Mar 2024 Surfaces, Manifolds and Related Topics, *Université du Luxembourg*, Luxembourg

Nov 2023 Forum des jeunes mathématicien·nes, *Université Libre de Bruxelles*, Belgium

Oct 2023 Probability and Geometry in, on and of non-Euclidian spaces, *Centre International de Rencontres Mathématiques*, Marseille, France

Jun 2023 Interactions between geometry and topology, *Banyuls de la Marenda*, France

Mar 2023 Geometry, probability and their synergies, *Les Diablerets*, Switzerland

Nov 2022 Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique, *Institut Mathématique d'Orsay*, France

Sep 2022 Workshop on Topological Methods in Geometry, *Centre de Recerca Matemàtica*, Barcelona, Spain

Mar 2020 Bernardo Cascales Functional Analysis School, *Universidad de La Laguna*, Spain

Academic References

Gérard Besson
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